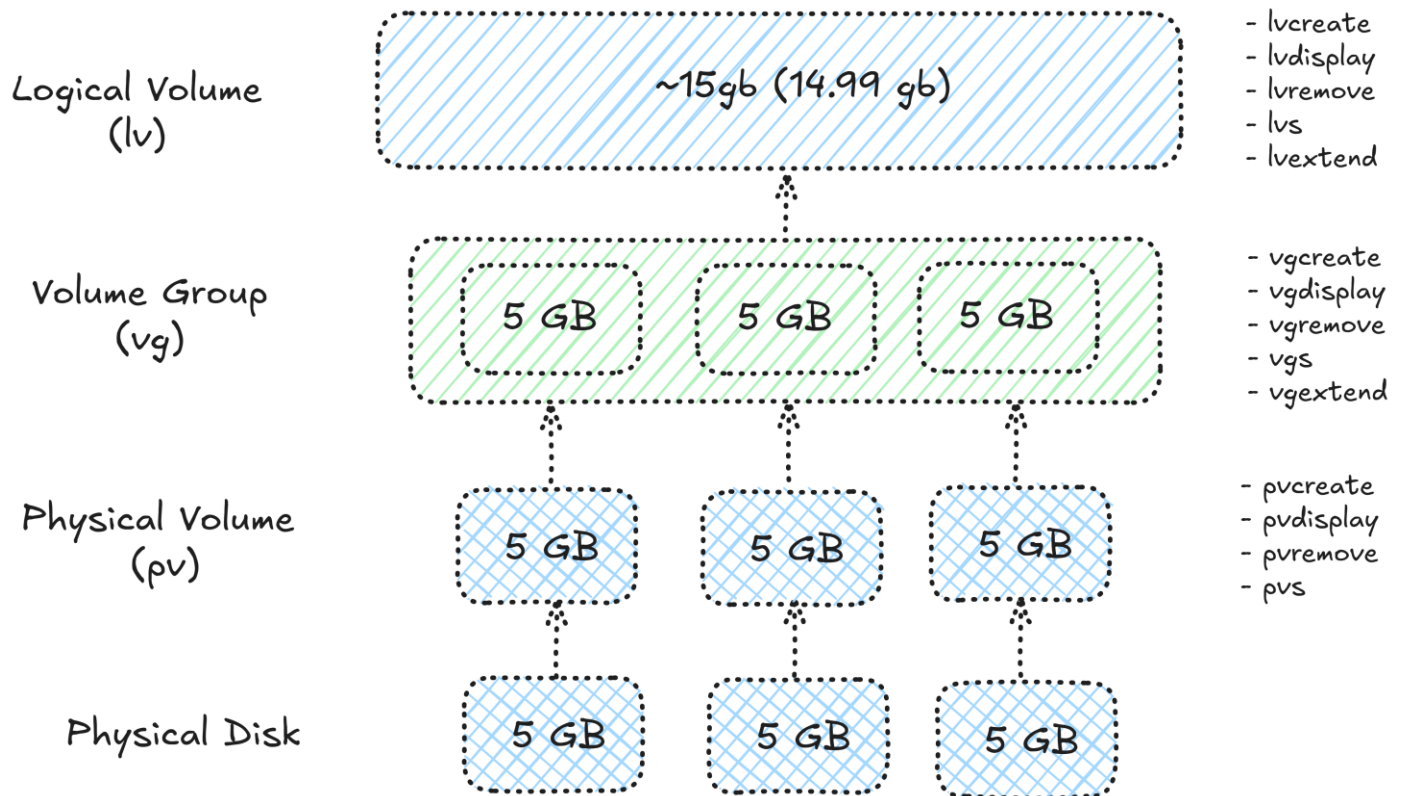


Task – add a user with your last name and make this user equivalent to root user.

Storage management:

- You can add a new partition, if required this partition can be extended. This technique is basically called as Logical Volume Manager (LVM).
- In LVM, you take various HDDs (of same size or different sizes) and logically club them together. Later if required, this logical volume can be extended or it can be shrunk.



Steps for creating an LVM:

1. Attach 2/3 disks to the linux VM.
2. Create partitions using “**fdisk**” utility.
3. Create a physical volume using “**pvcreate**” command.
4. Create a volume group using “**vgcreate**” command.
5. Create a logical volume using “**lvcreate**” command.
6. Create a file system for this newly created logical volume using “**mkfs**” command.
7. Create a new mount point using “**mkdir**” command.
8. Mount this partition
 - Temporary (using **mount** command) → until next reboot. Means after reboot, it will be gone.
 - Permanent (using **/etc/fstab** file) → it will be there even after reboot or shutdown.
9. Verify.

Commands for storage management in linux:

- ✓ “**df -h**” → this command displays all the mounted partitions that are in use.
- ✓ “**lsblk**” → this command displays all the block (hardware) devices connected to linux.
- ✓ “**mount**” → this command mounts a filesystem on linux.
- ✓ “**umount**” → this command unmounts a file system in linux.
- ✓ “**du -sh folder_name**” → this command shows the size of a folder in human readable format.

Naming convention of HDD and partitions:

sda = SCSI Disk A

hda = Hard Disk A



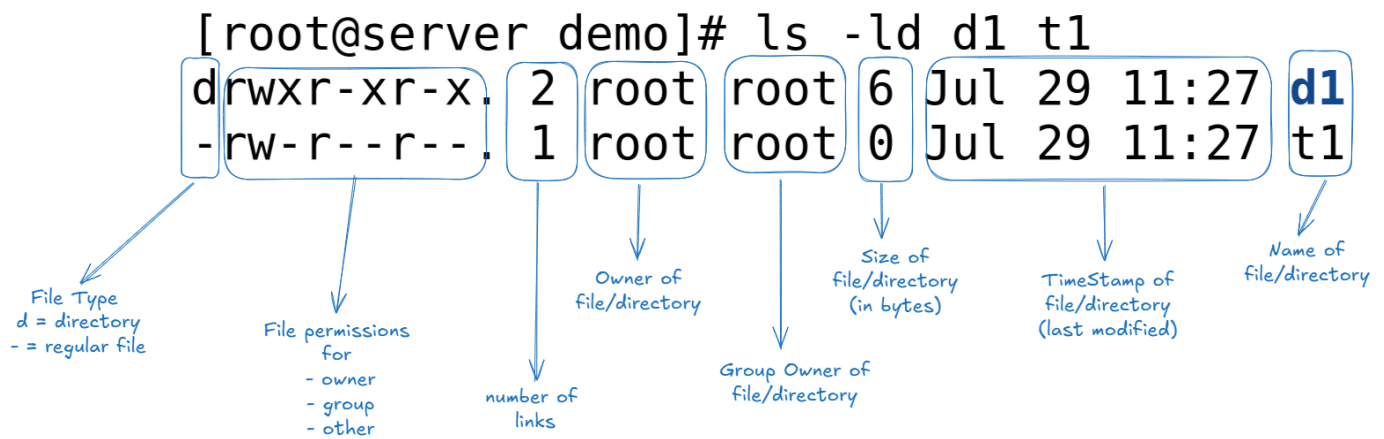
Task – add 3 disks of 10 GB each. And create 2 LVMs out of it. Size of 1st LVM will be 15gb and size of 2nd LVM must be 15gb. 1st LVM must have EXT4 file system and 2nd LVM must have XFS file system mounted on /lvm1 and /lvm2 respectively.

Permissions

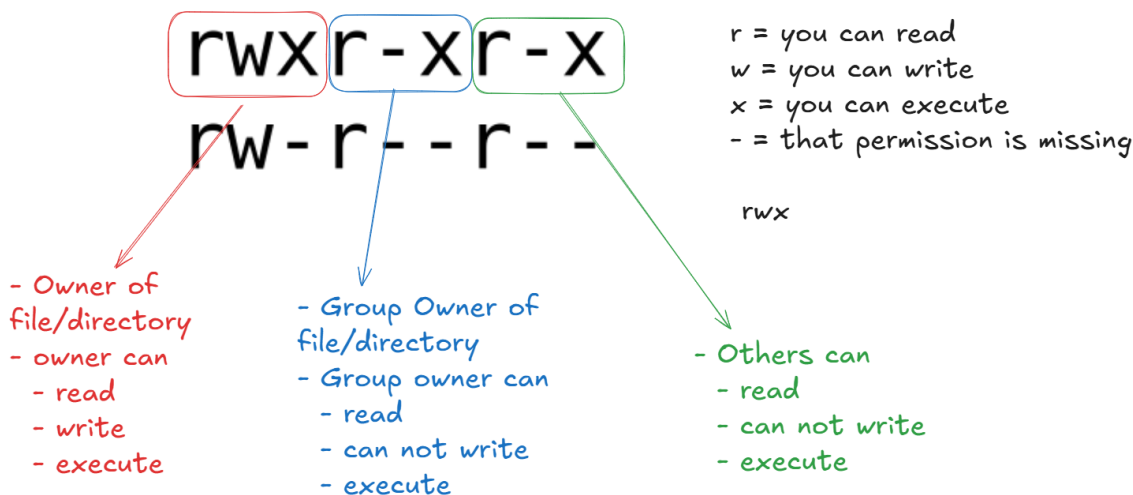
In linux, we have permissions like: - Read (r) - 4 - Write (w) - 2 - Execute (x) - 1 - ----- - Total - 7 - -----	In linux, permissions are applied on: - User/owner (u) - Group (g) - Others (o) - ----- - All (a) - -----
--	---

Commands for permissions

- "ls -l" → lists all the permissions of a file or directory
- "chmod" → this command changes the permissions for a file or directory.
- "chown" → this command changes the owner of a file or directory.
- "chgrp" → this command changes the group ownership of a file or directory.



Elaborating permissions:



Permissions can be applied in 2 ways in linux:

1. Numerical method
 - a. Here we only use numbers to change the permissions.
 - b. # **chmod 640 t1.txt**
2. Symbolical/alphabetical method
 - a. We use symbols (rwx, ugo, a, +=-)

*Task – for file t3, give **all** permissions to **all the users** and then remove 'write' permission from **others** using symbolical method.*

Owner and group ownership:

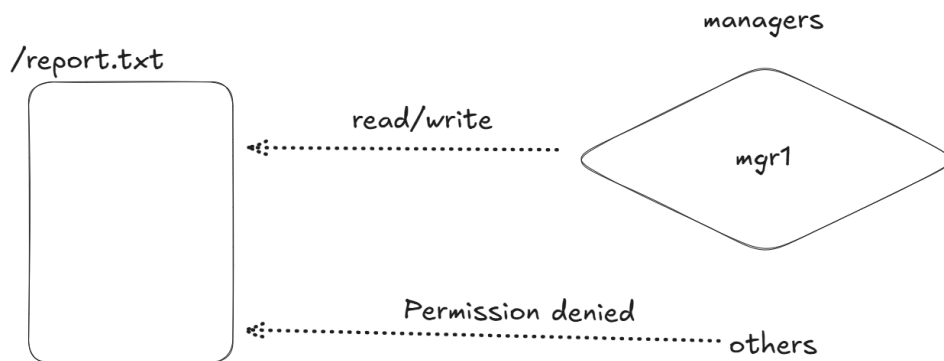
- ✓ By default, owner and group owners are same. Whoever create a file or directory, becomes owner of the file and becomes group owner of that file.
- ✓ To change the owner, command is “chown”.
 - Syntax: **chown** <owner name> <file/directory name>
- ✓ To change the permissions for a group owner, use command “chgrp or chown”
 - Syntax: **chgrp** <group owner name> <file/directory name> OR
 - Syntax: **chown** :< group owner name> <file/directory name>

Create a group

groupadd <groupname>

Task

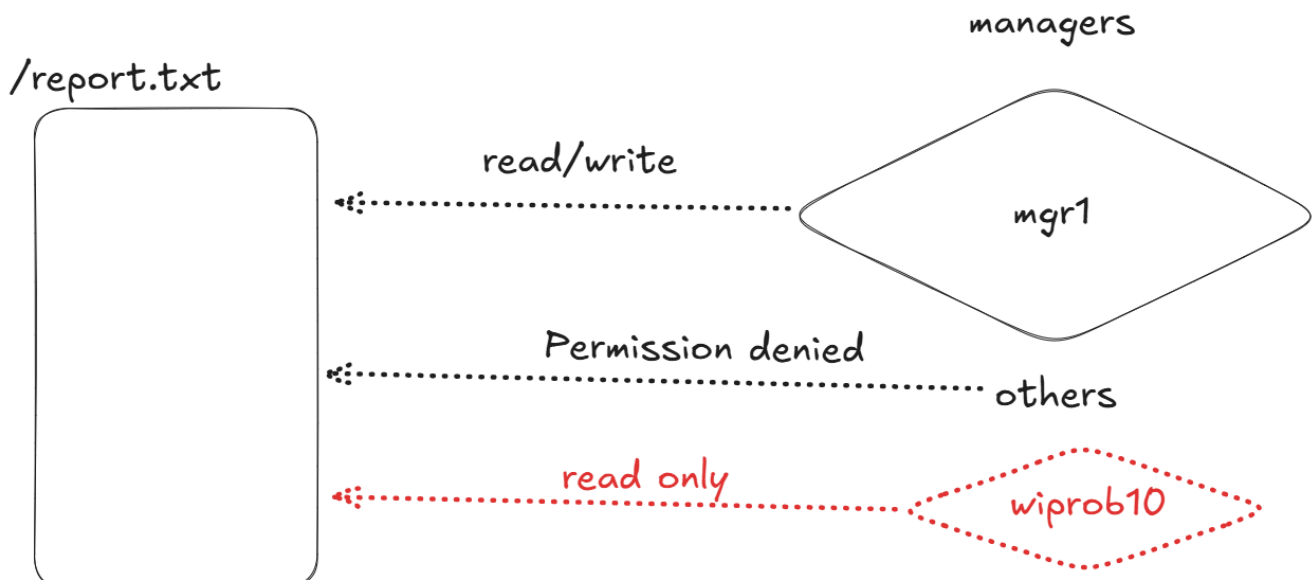
Create a file “/report.txt” and allow only managers group (having mgr1 user in it) to read and write into this file. Anyone else tries to access it must get “permission denied” message.



Solution

- Create a group “managers”.
- Create a user (mgr1) and add it to this group.
- Create a file “/report.txt”.
- Change the permissions.

Task – add another group (Wiprob10) to previous task and provide it “read-only access”.



ACL

- It is the extended form of Linux RWX permissions
- Stands for Access Control List.
- ACL is helpful in providing *fine-grained* access to any resource (file or directory).
- Commands to work with ACL:
 - **Getfacl** → to list ACL is applied on file/directory or not.
 - ✓ Syntax: `getfacl <file-path>`
 - **Setfacl** → set/change the ACL on a file/directory.
 - ✓ Syntax 1 (user): # `setfacl -m u:username:permissions <path-of-file/directory>`
 - ✓ Syntax 2 (group): # `setfacl -m g:groupname:permissions <path-of-file/directory>`

Package management:

- In windows, we have software and in Linux, we have PACKAGES.
- Packages are installed, update or removed using a package manager.
 - RedHat → RPM, YUM, DNF
 - CentOS → RPM, YUM, DNF
 - Ubuntu → APT, DPKG, DEB
 - Kali / Back Track → APT, DPKG, DEB
 - Suse linux → YaST
 - Windows → NuGet, Chocolatey, PowerShell
- Naming convention for packages in linux
 - **Name-version-release-architecture.rpm**
 - ✓ Linux packages support 3 types of architecture:
 - a. "i686" → 32-bit system
 - b. "x86_64" → 64-bit system
 - c. "noarch" → no architecture (for both 32 & 64 bit)
- **RPM**
 - Redhat Package Manager
 - RPM is default package manager in Redhat/Centos operating system.
 - RPM can
 - ✓ Install: # **rpm -ivh <full-package-name>.rpm**
 - a. "i" = Install
 - b. "v" = Verbose (show background processes on terminal)
 - c. "h" = hashing (completion bar)
 - ✓ Update: # **rpm -Uvh <full-package-name>.rpm**
 - a. "U" = Update
 - ✓ Query/Information: # **rpm -q <full-package-name>**
 - a. "q" = query
 - ✓ Remove: # **rpm -e <full-package-name>**
 - a. "e" = erase
- **YUM**
 - Yellowdog update modifier
 - This package manager downloads and install the packages from internet repositories.
 - ✓ RedHat – RedHat Satellite Server (subscription-based repo)
 - ✓ Centos – Online Mirror servers (free)
 - YUM is useful for installing packages with its dependencies.
 - YUM can
 - ✓ Install: # **yum install < package-name>**
 - ✓ Update: # **yum update < package-name>**
 - ✓ Query/Information: # **yum info < package-name>**
 - ✓ Remove: # **yum remove < package-name>**

Process management:

Commands:

- ✓ # **top** → Table Of Process → equivalent to task manager in windows.
- ✓ # **ps tree** → process tree → hierarchy of the processes.
- ✓ # **ps -aux** → list all processes running under all the users on all terminals.
- ✓ # **kill -15 <process-id>** → kill a process properly by terminating all the process instances
- ✓ # **kill -9 <process-id>** → kill a process, there might issues with killing a process.

To check, which user is logged-in to the linux machine:

- #who
- #w